Visual Pathway

Vision involves structure concerned with reception, transmission and perception of visual impulses.

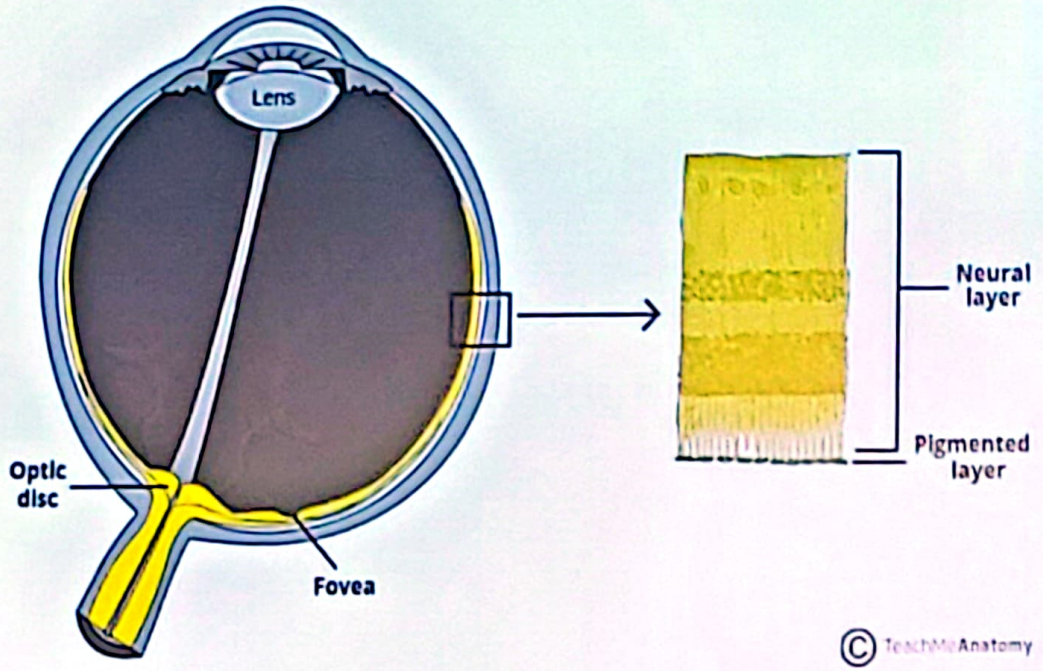
Structure concerned with pupil and reflexes are also of important consideration.

Visual pathway structure; Include, retina, optic nerve, optic chiasm, optic tract (medial and lateral roots), lateral geniculate body, optic radiation and visual area.

Retina: Delicate innermost membrane attached to basal lamina of choroid and in contact with inner vitreous body. Consist of 3 layers; Receptor cell layer of rod and cone;

Intermediate layer of bipolar cell; Ganglionic cell layer; axon converge at optic disc piercing the sclera to form optic nerve. The area devoid of the rod and cone cells (blind spot).

Optic nerve: Responsible for sight, composed of axons of cells in the ganglionic layer of retina emerging at the posterior-medial part of eyeball. It runs backward and medially piercing through optic canal to enter middle cranial fossa where it joins optic chiasma. It is about 4 cm long. 25mm of it is intra-orbital, 5mm intra-canalicular, 10mm intracranial enclosed in 3 meninges.

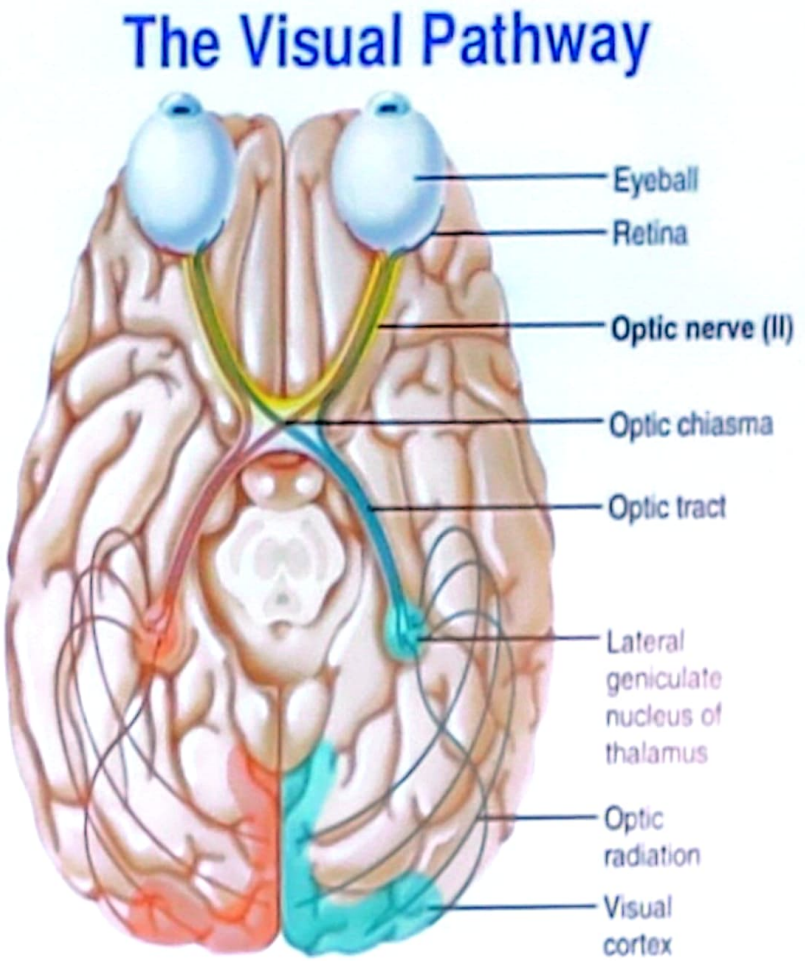


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Optic chiasma: Formed by the decussating fibres of optic nerve.

Fibres from the medial half of the retina (nasal fibres) cross the midline and enter the opposite optic tract.

Fibres from lateral half of the retina (temporal fibres) pass through the chiasma to enter the optic tract of the same side.



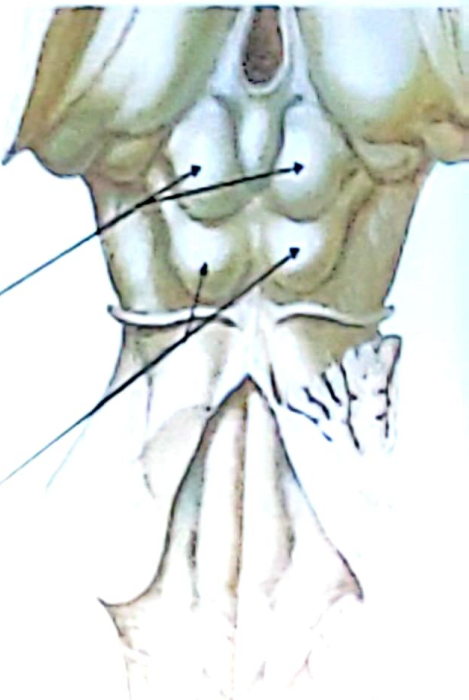
Optic tract: Each winds round the cerebral peduncle of the midbrain. Near the lateral geniculate body it divides into lateral and medial roots. The lateral root is thick and terminates in the lateral geniculate body. A few of its fibres bypass geniculate body to enter the superior colliculus, the pretectal nucleus and the hypothalamus. The medial root contains supraoptic commissural fibres of same side and nasal fibres of opposite side.

posterior surface

1. Four colliculi

These are rounded eminences that are divided by a vertical and a transverse groove into :

- Superior colliculi : are centers for visual reflexes
- Inferior colliculi : are lower auditory centers



Lateral geniculate ganglion: Receives lateral root of optic tract. Medially it connects the superior colliculus and laterally it gives rise to optic radiation.

The cells in lateral geniculate layer is made up of 6 layers; 2,3,5 receives ipsilateral fibres and layers 1,4,6 receives contralateral fibres.

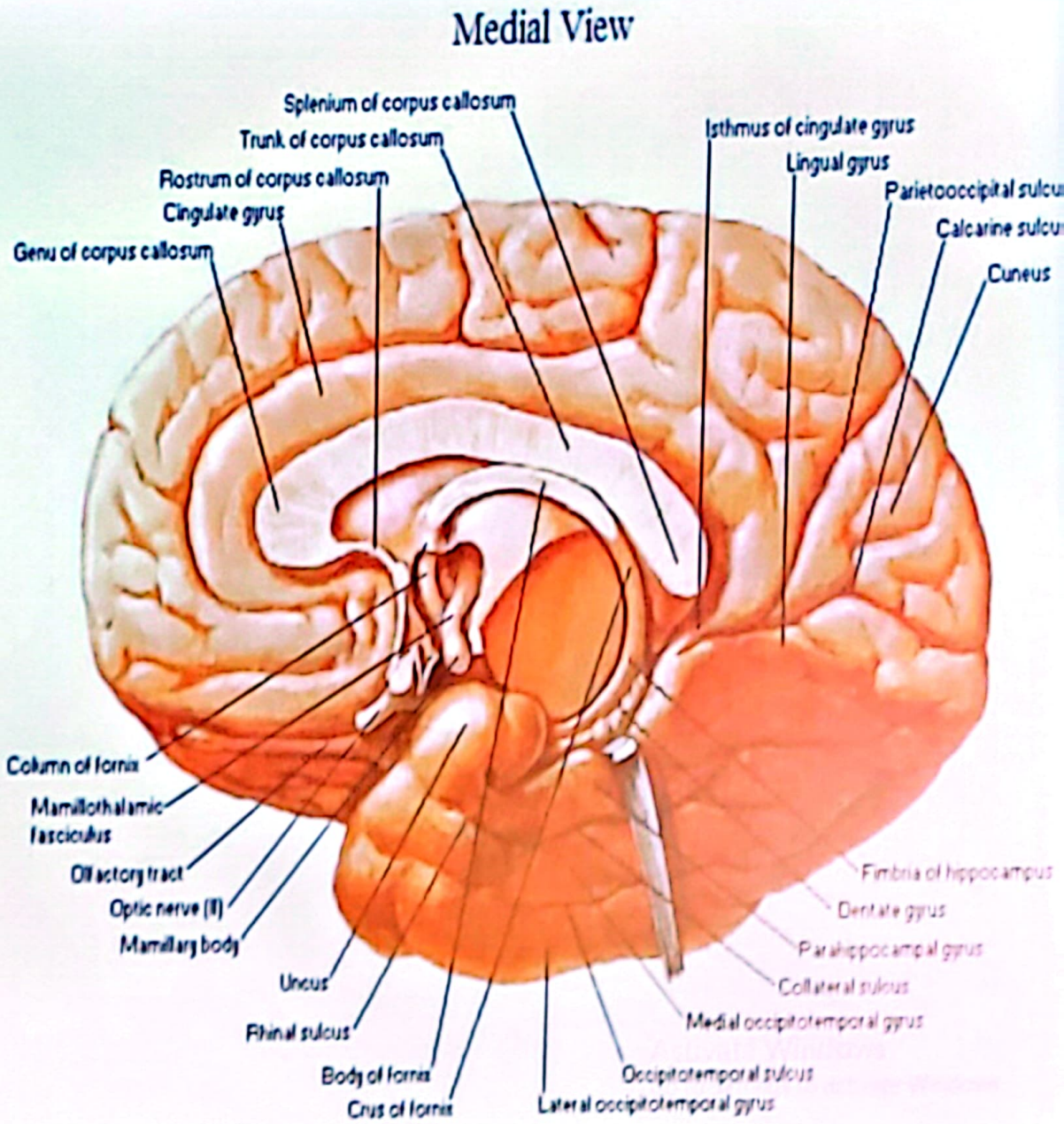
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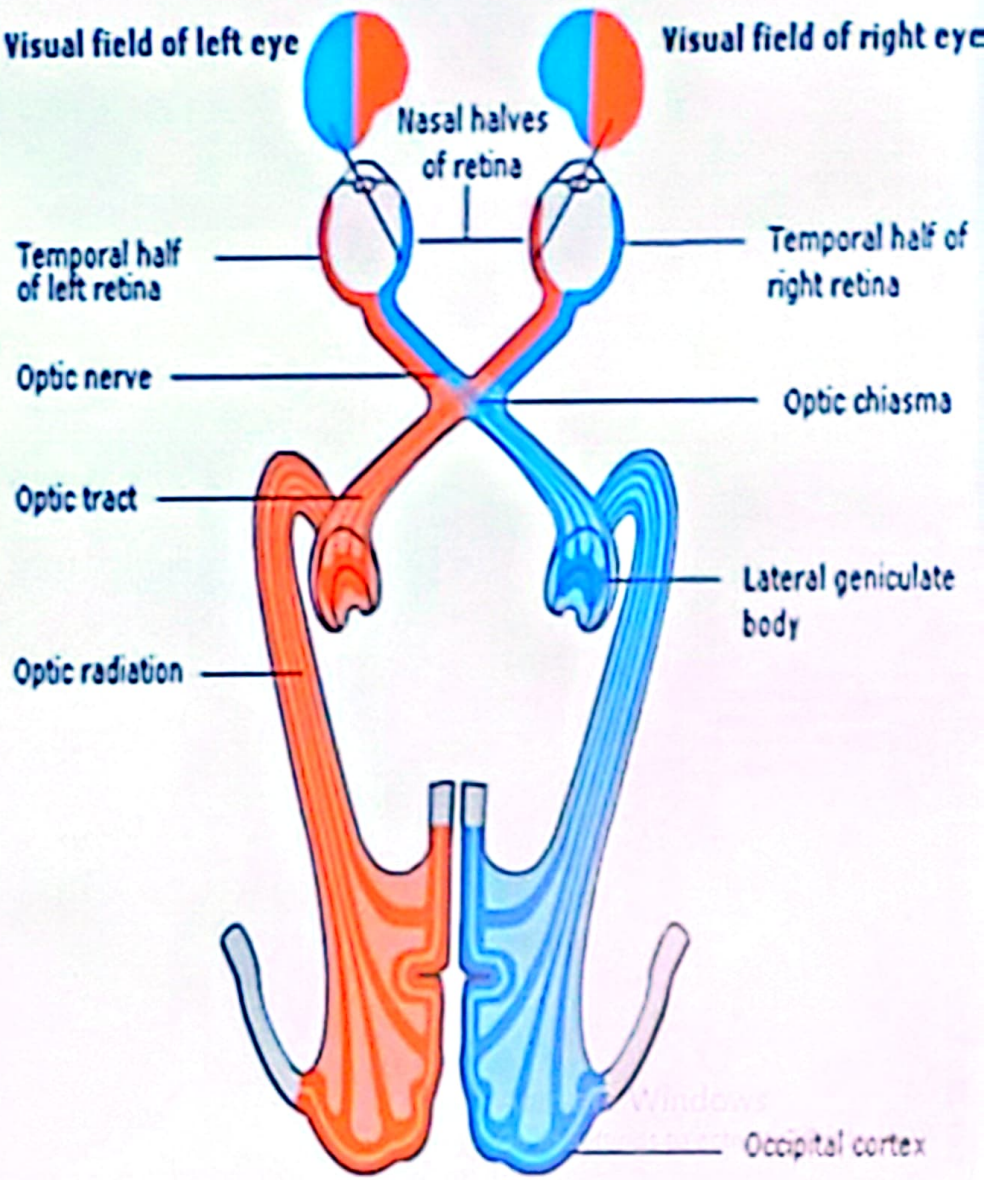
Optic radiation: Optic radiation begins from the lateral geniculate body. It first sweeps laterally through the retrolentiform part of the internal capsule in the temporal lobe before passing back to the visual cortex, striate area surrounding the calcarine fissure. It terminates in such a way that the upper and lower halves of retina are represented on the upper and lower lips of calcarine fissure respectively.



Visual cortex: Information pass through optic radiation to the striate area (17) of visual cortex where colour, size, shape, motion, illustration and transparency are appreciated separately.

Objects are identified by integration of these perceptions with past experience stored in the parastriate and peristriate areas (18,19).

Small proportion of fibres in the optic tract, sub-serving pupillary, ocular and head and neck reflexes bypass the geniculate body to reach superior colliculus and pretectal nucleus of midbrain.



From the pretectal nucleus fibres connect with Edinger-Westphal nucleus of both sides. The pathway explains the direct and consensual papillary response.

Pathway of light reflex: Light directed on the eye causes constriction of the pupil. It's mediated through retina, optic nerve, optic chiasma, optic tract, lateral geniculate body, pretectal nucleus the Edinger-Wesphal nucleus of C 3rd, the 3rd cranial nerve, ciliary ganglion, short ciliary nerves and constrictor pupillae.

Throwing light on one eye produces constriction of the pupil in both eyes (concensual reflex). It is due to the bilateral connections of the pretectal nucleus with Edinger-Wesphal nucleus.

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Pathway of accommodation reflex: Constriction of the pupil also takes place when looking at a near object. It is mediated through retina, optic nerve, the chiasma, optic tract, lateral geniculate body, optic radiation, visual area of cortex, superior longitudinal association tract, frontal eye field, 3rd nerve nucleus, 3rd cranial nerve, ciliary ganglion and ciliary muscle and sphincter pupillae muscles. It precludes pretectal nucleus.

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Clinical considerations:

i) Argyll-Roberson pupil: it is thought to be within the vicinity of the prectectal area. The light reflex is lost but the pupil contracts on accommodation.

ii) Lesion of retina or optic nerve will result to ipsilateral blindness in the affected segment but lesions of the optic tract and central parts of visual pathway will lead to contralateral homonymous defect.

iii) Lesion of optic chiasma will lead to bitemporal hemianopia- loss of vision in both temporal eye-fields.

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